MILITARY SPECIFICATION

COUPLER GROUP, ANTENNA OA-9123/SRC

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a shipboard tuned antenna coupler group (multicoupler) hereinafter referred to as the multicoupler, used to couple a group of transmitters, receivers, transceivers, or combinations thereof, to a common antenna for operation in the frequency range of 225.000 megahertz (MHz) to 399.975 MHz.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts), Packaging Of
MIL-R-24182	Radio Sets AN/SRC-20A And AN/SRC-21A (Including Radio Set AN/URC-9A And Radio Set Control C-3866/SRC)
MIL-C-24219	Coupler, Antenna, AN/SRA-33, Four-Channel, Automatically Tuned
MIL-C-28838	Communications Sets, UHF AN/WSC-3(V) Satellite And LOS Versions

STANDARDS

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, Specifications, Standards, And In Technical Documents
MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-129	Marking For Shipment And Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5985

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MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-965	Parts Control Program
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
DoD-STD-1686	Electrostatic Discharge Control Program For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)

HANDBOOKS

MILITARY

MIL-HDBK-241	Design Guide For Electromagnetic Interference (EMI) Reduction In Power Supplies

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 29, Part 1910, Subpart Z	Toxic And Hazardous Substances

(Copies of this Federal Regulation required by contractors in connection with specific procurement functions may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536D	Process Specification Procedures And Requirements For Preparation And Soldering Of Electrical Connections

(Copies of specifications, standards, handbooks, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS 310-C-1977	Racks, Panels, And Associated Equipment

(Application for copies should be addressed to the Electronic Industries Association, Engineering Department, 2001 Eye Street, NW, Washington, DC 20006.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The multicoupler shall conform to the requirements of MIL-E-16400, to the extent specified herein. The multicoupler shall be capable of replacing the AN/SRA-33 antenna coupler group specified in MIL-C-24219; and as a minimum the multicoupler shall be control interface compatible and interoperable with the AN/SRC-20( )/21 specified in MIL-R-24182, and AN/WSC-3(V) family radio equipment specified in MIL-C-28838. The functional description and the method of approach in the design of the OA-9123/SRC( ) antenna coupler group are specified in APPENDIX A.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Equipment composition. The multicoupler shall consist of bandpass filters, power supply, combiner, equipment rack or cabinet, cable assemblies, mating connectors, and all other related components, to allow the multicoupler to perform as specified herein. Special tools necessary for alignment, maintenance, and so forth, shall be included as part of the equipment (see 6.7).

3.3.1 Filter combining and interchangeability. The multicoupler shall be designed so that each filter is two-way interchangeable with other filters within the multicoupler, and filters may be inserted in the multicoupler to provide any number of channels from one to four, inclusive. Filter interchangeability shall be accomplished within the limits of maintainability requirements specified in 3.13. If any one, two, or three filter sections are removed, the multicoupler shall still be capable of operation without degradation of performance specified herein.

3.3.2 Test measurement and diagnostic equipment (TMDE). The multicoupler shall be capable of detecting, displaying, and isolating failures (see 6.4.5).

3.3.2.1 Test provisions. The multicoupler shall contain the test provisions specified in a through c:

a. Class A test provisions. Class A test provisions shall provide a means to verify that the on-line equipment (see 6.4.2) is operating properly. Class A provisions shall be accomplished by the use of built-in test (BIT) (see 6.4.4), built-in test equipment (BITE) (see 6.4.3), or both, and shall have the capability to detect at least 96 percent of all equipment failures.

b. Class B test provisions. Class B test provisions shall provide a means to isolate at least 94 percent of the equipment failures to the subassembly (see 6.4.6) level as specified in 1 and 2:

1. At least 84.6 percent of the equipment failures shall be isolated to a single subassembly.

2. The remaining equipment failures (no more than 11.4 percent of total failures) shall be isolated to three or fewer subassemblies. Class B test provisions shall be accomplished when the equipment is off-line (see 6.4.1), on-line, or both, through the use of BIT and BITE.

c. Class C test provisions. Class C test provisions shall provide a means to fault locate to electrical functions (that is, integrated circuit (IC), filter, mixer, voltage controlled oscillator (VCO), and so forth) and to align the system (see 6.4.9), unit (see 6.4.8), assembly (see 6.4.7), or subassembly (see 6.4.6). Class C provisions may be accomplished by the use of test points in conjunction with automatic test equipment or general purpose electronic test equipment (GPETE) and shall have the capability to isolate all the failures to the electrical functional items on a subassembly.

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3.3.2.2 BIT and BITE criteria. BIT and BITE criteria shall be as specified in a through c:

a. Protection from BIT and BITE failures. The prime equipment function shall not be impaired by the malfunction of BIT and BITE.

b. BIT and BITE indicator. BIT and BITE indicators shall be lights, meters, or other indicating devices installed in the equipment. It shall provide a GO or NO-GO indication at the level which it is intended and shall be made visible by removal of no more than one cover plate if it is covered. BIT and BITE indicators shall be located as specified in 1 through 3:

1. Class A test provision: On the control panel of the multicoupler group
2. Class B test provision: On the multicoupler group control panel
3. Class C test provision: On the subassembly

c. BIT and BITE activation. BIT and BITE activation shall be as specified in 1 and 2:

1. BIT and BITE used in Class A and B test provisions shall be energized automatically without operator's initiation.
2. BIT and BITE used in Class C test provisions may be energized automatically or manually. When a manually energized test mode is to be employed, a pushbutton switch shall be incorporated.

3.3.2.3 Test point criteria. Test points shall provide the requirements specified in a through d:

a. Signal quality and circuit performance: Test points, test jacks, or both, shall be provided to permit the injection of signals and the monitoring of signals at the input/output (I/O) terminals of the subassembly.

b. Fault isolation: Test points, test jacks, or both, shall be provided to permit the injection or measurement of signals at I/O terminals of the subassembly to isolate all faults to the electrical functions (that is, IC, filter, mixer, VCO, and so forth). Test points and test jacks shall provide means for performing fault isolation on the subassembly with the subassembly removed from the prime equipment.

c. Alignment: Test points, test jacks, or both, shall be provided to permit alignment of the equipment and equipment assemblies and subassemblies.

d. Calibration: Class A, B, and C test provisions shall be provided with test points, test jacks, or both, to permit calibration of the BIT and BITE functions of the equipment. Calibration of the Class A, B, and C BIT and BITE functions shall not exceed 2 hours.

3.2.2.4 GPETE. GPETE to be used at all levels of maintenance shall be selected from the standard and standard replacement equipments listed in MIL-STD-1364.

3.3.3 Power supply requirements. Design for low voltage supplies delivering up to 5 kilowatts (kW) at 300 volts direct current (VDC), or less, shall comply with the requirements in a through j:

a. The power density shall be output power divided by power supply envelope volume, including electromagnetic interference (EMI) filtering where required. Power density exceeding 2 watts per cubic inch shall require the approval of the procuring activity.

b. Junction temperature of semiconductor and microelectronic devices (diodes, transistors, hybrid ICs, and so forth) shall not exceed +110° Celsius (C) under worst-case conditions.

c. Case/hot spot temperatures shall not exceed 40°C rise above ambient temperature with a maximum temperature of +110°C for parts less than or equal to 3 watts (W) of dissipation; 55°C rise above ambient with a maximum temperature of +125°C for parts greater than 3 W of dissipation; 30°C rise above part ambient with a maximum temperature of +100°C for transformers; and 10°C rise above ambient, due to self-heating, with a maximum temperature of 85°C for capacitors.

d. Power supplies shall be designed to be maintainable at the piece part level; power supplies shall not be limited by design to repair by the manufacturer.

e. Power supplies shall not be encapsulated or embedded (potted) unless it can be shown by test to be necessary for heat removal or dissipation. This requirement shall not exclude conformal coating.

f. Power supply manufacturing shall include random vibration and temperature cycling of every unit under full electrical load; this may be done in the end item but shall be performed on a unit basis for spares.

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g. EMI requirements shall be tailored for the power supply which will enable the end equipments to conform to the class of MIL-STD-461 required by its specification. EMI specifications shall be developed for the power supply which will permit the power supply to be purchased separately from the end equipment.

h. The specification mean-time-beween-failures (MTBF) requirement shall be not less than 40,000 hours for Naval sheltered equipment at 55°C.

i. Power supplies shall be designed to withstand, and shall not be damaged by, load between an open circuit and a short circuit.

j. Power supplies shall be designed with a reserve load capacity of 20 percent.

3.4 Characteristics. Characteristics shall be as specified in 3.4.1 through 3.4.2.6.

3.4.1 Dimensions and weight. A complete four-channel multicoupler shall be designed for mounting in a standard 19-inch (in.) open rack as specified in RS 310-C-1977. The multicoupler shall not exceed 88.90 centimeters (cm) (35 in.) high, 76.20 cm (30 in.) deep, and shall weigh no more than 136.36 kilograms (kg) (300 pounds (lb)).

3.4.2 Environmental service conditions. The multicoupler shall conform to the operating requirements specified in 3.6 through 3.6.13 without degradation under any combination of environmental conditions except as specified in 3.4.2.1 through 3.4.2.6.

3.4.2.1 Shock and vibration. The shock requirements for the multicoupler shall be Grade A, Class II, lightweight, Type A of MIL-S-901. The multicoupler shall conform to the vibration requirements of MIL-STD-167-1, Type I.

3.4.2.2 Temperature. The temperature range of the multicoupler shall be as specified in the Non-operating temperature ranges paragraph for Range 4 of MIL-E-16400.

3.4.2.3 Humidity. The multicoupler shall maintain the specified performance when exposed to the conditions of the Humidity paragraph of MIL-E-16400.

3.4.2.4 Salt fog (spray). The multicoupler shall be capable of withstanding without damage the effects of the Salt fog test paragraph of MIL-E-16400 for sheltered equipment.

3.4.2.5 Fungus. The multicoupler shall conform to the Fungus-inert and fungus nutrient paragraph of MIL-E-16400.

3.4.2.6 Inclination. The multicoupler shall conform to the Inclination paragraph of MIL-E-16400.

3.5 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the Parts, materials, and processes paragraph of MIL-E-16400, except as specified in 3.5.1 through 3.5.4.1.

3.5.1 Parts control. The parts to be incorporated in the multicoupler shall be in accordance with MIL-STD-965, Procedure I.

3.5.2 Derating. Application and derating of electronic parts and materials shall be in accordance with APPENDIX B.

3.5.3 Materials. Materials shall be in accordance with 3.5.3.1 through 3.5.3.4.

3.5.3.1 Material flammability. Material used as part of the multicoupler shall conform to the Flammable materials paragraph of MIL-E-16400.

3.5.3.2 Unacceptable materials. The Unacceptable materials paragraph of MIL-E-16400 shall apply.

3.5.3.3 Carcinogens. Materials identified as carcinogens (cancer producing substances) in the Code of Federal Regulations, Title 29, Part 1910, Subpart Z shall not be used as part of the multicouplers.

3.5.3.4 Material toxicity. Materials shall not be used in the multicoupler which, when exposed to a temperature range of -65°C to +125°C, emit air contaminants in excess of the threshold limit values specified in the Code of Federal Regulations, Title 29, Part 1910, Subpart Z.

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3.5.4 Processes. Processes, such as casting, corrosion protection and corrosion resisting treatments, painting, welding (structural), welds, and resistance (electrical interconnections) shall be in accordance with the Processes paragraph of MIL-E-16400. Internal aluminum surfaces which are not to be painted shall be treated for protection in the same manner as aluminum surfaces which are to be painted.

3.5.4.1 Finish. The external finish shall be light gray enamel conforming to MIL-E-15090, Type III, Class 2.

3.6 Electrical design and construction. The design and construction of the multicoupler shall conform to the requirements of the Electrical design and construction paragraph of MIL-E-16400.

3.6.1 Electrical overload protection. Electrical overload protection shall conform to the Electrical overload protection paragraph of MIL-E-16400. All parts and circuits in the multicoupler which are likely to carry an overload due to failure or improper adjustments shall have protection devices, or shall be properly rated to carry an overload without permanent damage to the multicoupler.

3.6.2 Input power interface. The multicoupler shall be designed to operate from an alternating current (AC) power source of 115 volts (V), 60 hertz (Hz), single-phase, Type I in accordance with DoD-STD-1399, Section 300. The multicoupler shall comply with the Interface characteristics and constraints paragraph of DoD-STD-1399, Section 300.

3.6.3 Impedance. Nominal impedance exhibited to the antenna terminal and to each radio terminal shall be 50 ohms.

3.6.4 Radio frequency (RF) power. The multicoupler shall be capable of accepting RF power on all input filters simultaneously, with each filter able to pass a minimum signal of 100 W average carrier power amplitude modulation modulated 80 percent with a 1 kilohertz (kHz) tone.

3.6.5 Frequency range. Each multicoupler filter shall be capable of operating over the 225.000 MHz to 399.975 MHz frequency band.

3.6.6 Multicoupler mode control. The multicoupler shall have front panel control, providing for independent operation of each of the four filter sections in each of the modes specified in a through c:

a. Local
b. Preset
c. Remote

The local and preset modes shall not be overridden by the remote mode of operation.

3.6.6.1 Local control mode. In the local mode of operation, the multicoupler shall be capable of providing each of the four filter sections with independent quick-change selection of any of the 7000 frequencies in the 225.000 MHz to 399.975 MHz frequency band in 25-kHz increments. In the local mode, each filter section shall automatically tune to the selected frequency.

3.6.6.2 Preset control mode. In the preset mode of operation, the multicoupler shall be capable of having each of the four filters preset (pretuned) to 20 different frequency channels between 225.000 MHz and 399.975 MHz in 25-kHz increments. The time required to change all multicoupler presets (80 independent frequency channels) shall not exceed 20 minutes. The multicoupler shall be capable of having the same frequency preset in a given channel of each of the four filters simultaneously. The time required to change all multicoupler presets in this manner (20 independent frequency channels) shall not exceed 5 minutes. In the preset mode, each filter shall automatically tune to the selected frequency channel.

3.6.6.3 Remote control mode. In the remote control mode of operation, the multicoupler shall provide for each of the four filter sections being independently accessed from a remote location to allow operation with a companion AN/SRC-20( )/21( ) or AN/WSC-3(V) family radio by selecting one of the 20 preset frequency channels established in 3.6.6.2. In the remote mode, each filter section shall be automatically tuned to the selected preset frequency channel.

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3.6.6.4 Equipment protection. The multicoupler shall have a protection system so that whenever any radio utilizing the multicoupler is keyed, all other radios and their associated filter sections, using the same multicoupler which are tuned within ±3 MHz of the keyed radio, will be protected by disabling their respective transmitter keylines and disconnecting their RF power lines. This protection system shall not inhibit the multicoupler from having two or more filter sections tuned within ±3 MHz of one another to allow RF reception during unkeyed conditions (receive only) nor shall the protection system prevent any filter from tuning through any other preset frequency already established by any of the remaining filter sections under any conditions.

3.6.6.5 Frequency indicator. Each multicoupler filter section shall provide a local display to indicate the frequency to which the multicoupler is tuned.

3.6.6.6 Spontaneous frequency selection. No combination of service conditions (including environmental conditions specified herein) shall cause any multicoupler filter to spontaneously change frequency.

3.6.6.7 Tuning time. When a filter section is switched from one frequency to another frequency within the band, locally or remotely, that filter section shall be available for operation within 10 seconds of activating the frequency selection controls. When the multicoupler receives multiple requests for frequency change, locally or remotely, the multicopuler shall satisfy these requests on a first come first served basis.

3.6.7 Voltage standing wave ratio (VSWR). Maximum VSWR exhibited to a 50-ohm transmission line at the antenna terminal and at each radio terminal shall not exceed 2 to 1 for all frequencies from 225.000 MHz to 399.975 MHz.

3.6.8 Insertion loss. The total insertion loss of the multicoupler when configured with all channels (that is, between a filter section input port and the multicoupler output port), shall be no greater than 2.0 decibels (dB) over the multicoupler frequency range under standard conditions and shall be no greater than 3.0 dB under environmental service conditions. (The standard conditions are: 115 volts alternating current (VAC); 60 Hz, single-phase power; +20°C to +35°C ambient temperature; no shock, inclination, or vibration; ambient humidity and elevation.)

3.6.9 Selectivity. The response of the multicoupler filter(s) to signals ±150 kHz from the selected (center) frequency shall be attenuated no more than 3 dB, referenced to the response at the center frequency. The response of the filter(s) to signals ±5.0 MHz from the selected frequency shall be attenuated no less than 60 dB referenced to the response at the center frequency.

3.6.10 Duty cycle. The multicouoler shall be capable of continuous operation.

3.6.11 Radio set compatibility. The multicoupler shall be compatible in all respects (control and power) with the AN/SRC-20/20A, AN/SRC-21/21A, and AN/WSC-3(V)/6/7.

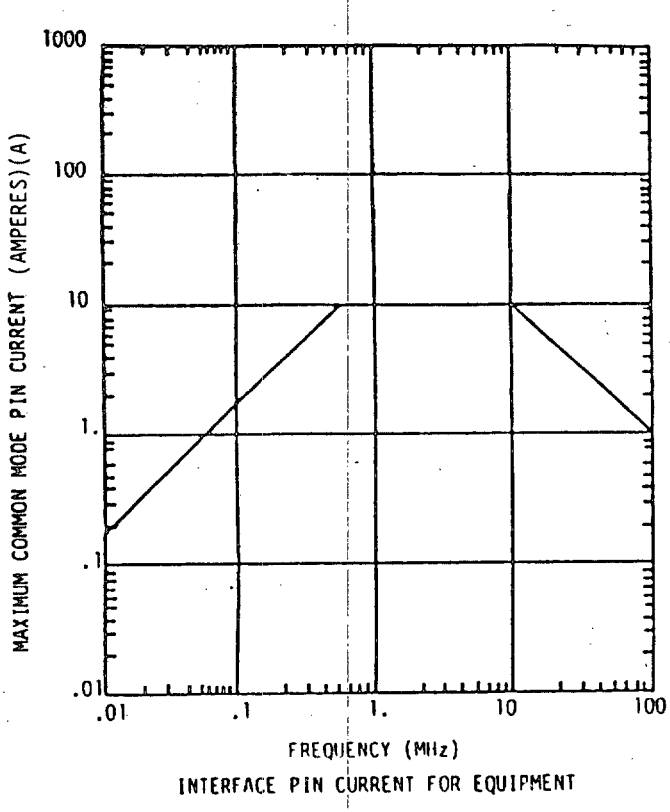
3.6.12 Electromagnetic environmental effects (E^3). E^3 shall conform to 3.6.12.1 through 3.6.12.2.1.

3.6.12.1 EMI. The multicoupler shall be self-compatible and shall conform to the requirements of MIL-STD-461 for Class A4 equipment. MIL-HDBK-241 may be used for power supply design guidance.

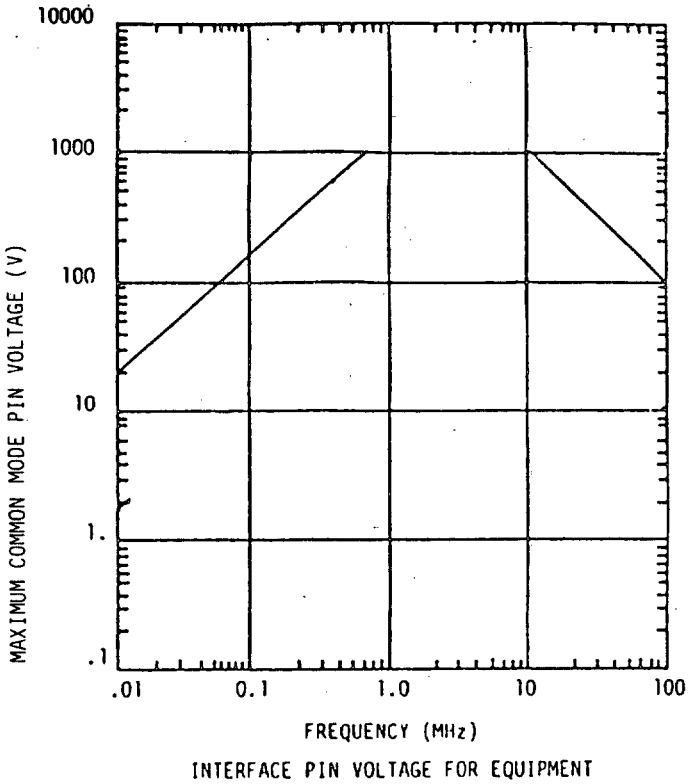
3.6.12.2 Electromagnetic pulse (EMP). The multicoupler shall be hardened or shall be intrinsically hard against the EMP resulting from an exo-atmospheric nuclear burst to survive the level specified in 3.6.12.2.1.

3.6.12.2.1 EMP test signal. All equipment and subsystem interface pins and terminals of leads, including grounds and neutrals shall not exhibit any permanent malfunction, degradation of performance, or deviation from specified indications beyond the tolerances, and recovery times, after being subjected to a test signal having either the waveform and common mode current level or the waveform and common voltage level both shown on FIGURE 1, whichever occurs first at the specific test frequency.

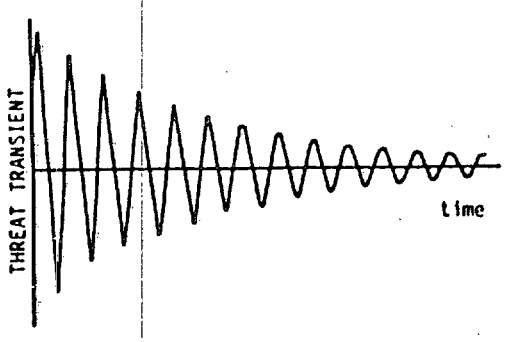
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INTERFACE PIN CURRENT FOR EQUIPMENT



INTERFACE PIN VOLTAGE FOR EQUIPMENT



$$x(t) = X(f_o) \sin(2\pi f_o t)\, e^{(-\pi f_o t/Q)}$$

$$Q = 16 \pm 4$$

FIGURE 1. Test signal waveform characteristics.

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3.6.13 Intermodulation. Intermodulation products detected at the antenna and a selected receiver termination shall be less than -40 decibels referred to one milliwatt (dBm) when the multicoupler is simultaneously processing the outputs of three transmitters and one receiver separated in frequency by 35 MHz.

3.7 Mechanical design and construction. The mechanical design and construction requirements shall be in accordance with 3.7.1 through 3.7.3.

3.7.1 Degree of enclosure. The multicoupler shall conform to the splashproof requirements of MIL-STD-108.

3.7.2 Controls, indicators, and panel layouts. The design of controls and indicators, and their arrangement on operator and maintenance panels, shall conform to the criteria of MIL-STD-1472 and shall be in conformance with the Controls, indicators, and panel layout paragraph of MIL-E-16400. The color coding and marking of pin jacks and other test points shall be in conformance with MIL-STD-415. Abbreviations used on panels shall conform to MIL-STD-12.

3.7.3 Magnetic requirements. The multicoupler shall conform to the Magnetic requirements paragraph of MIL-E-16400.

3.8 Thermal design. Thermal design shall conform to the Thermal design paragraphs of MIL-E-16400.

3.9 Identification and marking. Identification and marking of the multicoupler shall conform to the Identification and marking paragraph of MIL-E-16400.

3.10 Safety criteria. Multicoupler design shall conform to the Safety criteria paragraph of MIL-E-16400, except as specified in 3.10.1 through 3.10.4.1.

3.10.1 Leakage current. The leakage current of the multicoupler shall not exceed 5 milliamperes (mA) AC or direct current (DC).

3.10.2 Multicoupler power connections. Multicoupler power connectors for 115 VAC input power and interconnecting AC power cable connectors between equipments shall have only three pins designated A, B, and C. Pin assignments and color code shall be as shown in TABLE I.

TABLE I. Pin assignment and color code.

Designation pin	Conductor assignment	Color code
A	115 VAC return	White
B	Safety ground	Green
C	115 VAC high (hot)	Black

Black and white color code shall be maintained from the input connector pins to all components having the same voltage and frequency as the input power.

3.10.3 Single-phase hard-wired equipment, color coded conductors. When a single-phase multicoupler is hard-wired to a power source, the color code of the power conductors inside the multicoupler shall conform to 3.10.2.

3.10.4 Multicoupler electrical power ON-OFF switch. A switch for disconnecting the multicoupler from all electrical power systems shall be mounted on the multicoupler front panel and the function clearly labeled. The switch shall break all power conductors of the power circuit.

3.10.4.1 Power indicator lamp. A green power indicator lamp shall be mounted near the ON-OFF switch to indicate when the multicoupler is energized. The lamp shall be connected to the load side of the switch and across the input power conductors.

3.11 Electrostatic discharge (ESD). ESD protection shall be provided in accordance with the Design protection paragraph of DoD-STD-1686.

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3.11.1 ESD marking. Multicouplers containing parts that are sensitive to ESD shall be marked as specified in a through c:

a. Assemblies. Assemblies shall be marked in accordance with the Assemblies paragraph of DoD-STD-1686.

b. Equipment enclosures. Equipment enclosures shall be marked in accordance with the Equipment enclosures paragraph of DoD-STD-1686.

c. Electrostatic discharge sensitive (ESDS) nonstandard parts. ESDS nonstandard parts shall be marked with the sensitive electronic device symbol of MIL-STD-129.

3.12 Quantitative reliability requirements. The multicoupler, including BITE, shall have a lower test MTBF (θ_1 as defined in MIL-STD-781) of 1500 hours. The upper test MTBF (θ_0 as defined in MIL-STD-781) shall be 4500 hours.

3.13 Maintainability requirements. Maintainability requirements shall be in accordance with 3.13.1 and 3.13.2.

3.13.1 Quantitative corrective maintenance. The multicoupler, including BITE, shall have a mean-corrective-maintenance-time (M_{ct}), not exceeding 10 minutes and a maximum-corrective-maintenance-time (M_{maxct}) (95th percentile), not exceeding 30 minutes when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, printed circuit boards (PCBs)) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The M_{ct} includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective-maintenance actions. BITE circuits are included in the corrective-maintenance time requirement. The equipment is considered to be in a corrective-maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.13.2 Subassembly (module) maintainability. Repairable modules shall have an M_{ct} of 1.0 hours and an M_{maxct} of 2.0 hours (95th percentile) at the organizational, intermediate, or depot level maintenance as appropriate. M_{ct} for module repair includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks. For purposes of maintainability, a module is considered the lowest replaceable subassembly (printed circuit card) for which repair is accomplished by the replacement of piece parts.

3.14 Workmanship. Workmanship shall be in accordance with 3.14.1 and 3.14.2.

3.14.1 General workmanship. Workmanship shall conform to the Workmanship paragraph of MIL-E-16400. Soldering, soldering procedures, and soldering processes shall conform to WS 6536D.

3.14.2 Workmanship screen. All multicouplers shall withstand a defect detection vibration screen of random type vibration (see 4.8.2.1) at $0.04g^2$/Hz ±3 dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.8.2.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specifed in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality for the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (see 4.4.1)
   2. Production control inspection (see 4.4.2)
   3. Environmental inspection (see 4.4.3)
c. Inspection of preparation for delivery (see 4.9)

4.3 First article inspection. Unless otherwise specified (see 6.2), four units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

TABLE II. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Reliability qualification test	3.12	4.6	X			
Parts control	3.5.1	4.5	X			
Finish	3.5.4.1	4.5	X			
Material toxicity	3.5.3.4	4.5	X			
Material flammability	3.5.3.1	4.5	X			
Carcinogens	3.5.3.3	4.5	X			
TMDE	3.3.2	4.5	X			
Electrical overload protection	3.6.1	4.5	X			
Input impedance	3.6.3	4.5	X			
Radio set compatibility	3.6.11	4.5	X			
Multicoupler mode control	3.6.6	4.5	X			
E^3	3.6.12	4.5.7	X			X
Controls, indicators, and panel layouts	3.7.2	4.5	X			
Magnetic requirements	3.7.3	4.5	X			
Thermal design	3.8	4.5	X			
Identification and marking	3.9	4.5	X			
Safety criteria	3.10	4.5	X			
Multicoupler power connections	3.10.2	4.5	X			
Single-phase hard-wired equipment, color coded conductors	3.10.3	4.5	X			
Multicoupler electrical power ON-OFF switch	3.10.4	4.5	X			

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TABLE II. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Power indicator lamp	3.10.4.1	4.5	X			
Filter combining and interchangeability	3.3.1	4.5	X			
Dimensions and weight	3.4.1	4.5.1	X			
Shock and vibration	3.4.2.1	4.5	X			
Temperature	3.4.2.2	4.5	X			
Salt fog (spray)	3.4.2.4	4.5	X			
Fungus	3.4.2.5	4.5	X			
Inclination	3.4.2.6	4.5	X			
General workmanship	3.14.1	4.8.1	X	X		
Maintainability demonstration	3.13	4.7	X			
RF power	3.6.4	4.5.2	X		X	
Frequency range tuning and frequency interlock	3.6.5	4.5.3	X	X		
VSWR	3.6.7	4.5.4	X	X		
Insertion loss	3.6.8	4.5.5	X	X		
Selectivity	3.6.9	4.5.6	X	X		
Leakage current	3.10.1	4.5.8	X	X		
Input power interface	3.6.2	4.5.9	X		X	
Power and power factor	3.6.2	4.5.9	X			
Voltage and frequency tolerance	3.6.2	4.5.9	X			
Voltage and frequency transients	3.6.2	4.5.9	X			
Spike voltage	3.6.2	4.5.9	X			
Power interruption	3.6.2	4.5.9	X			
Harmonic current	3.6.2	4.5.9	X			
Operating test	--	4.5.10	X	X		
Local control mode	3.6.6.1	4.5	X	X		
Preset control mode	3.6.6.2	4.5	X	X		
Remote control mode	3.6.6.3	4.5	X	X		
Equipment protection	3.6.6.4	4.5.7	X	X		
Frequency indication	3.6.6.5	4.5.7	X	X		
Tuning time	3.6.6.7	4.5	X	X		
Degree of enclosure 1/	3.7.1	4.5	X		X	
Humidity	3.4.2.3	4.5	X			X
Workmanship screen	3.14.2	4.8.2	X	X		
Leakage current 1/	3.10.1	4.5.8			X	

1/ Only one multicoupler from every other lot shall be subjected to the enclosure and leakage tests.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each multicoupler offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the tests specified in Group A of TABLE II. General workmanship and workmanship screen shall be the first two inspections performed.

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4.4.2 Production control inspection (Group B). Production control inspection, including sampling, shall consist of the tests specified in Group B of TABLE II, and the inspection procedures of MIL-STD-105, using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. The acceptable quality level shall be 6.5 percent defective. Production control inspection shall be performed on multicouplers that have passed production inspection. The multicouplers shall satisfactorily conform to the requirements of production control inspection prior to release for shipment.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall consist of the tests specified in Group C of TABLE II and shall be conducted on multicouplers which have been subjected to, and have passed, the production inspection.

4.4.3.1 Sampling for environmental inspection. One multicoupler shall be selected from each successive 100 multicouplers, or fraction thereof, produced. The first sample multicoupler shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.4 Reinspection of conforming environmental sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to, and have passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Test methods. The examinations and tests specified shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.5.1 through 4.5.8.2.

4.5.1 Dimensions and weight. In addition to measuring and weighing the complete multicoupler, individual major assemblies (see 3.3) shall be measured and weighed separately.

4.5.2 RF power. Each multicoupler shall be operated continuously at full RF power (100 W average nominal input to each filter section for a period not less than 1 hour). I/O power shall be recorded throughout the test. The test frequency shall be at the low end of the frequency range. At the end of the 1-hour period, each section shall be operated for 5 minutes at a frequency in the middle of the frequency range (about 300.000 MHz) and then for 5 minutes at a frequency at the high end of the frequency range, at full power. There shall be no evidence of excessive heating or malfunction.

4.5.3 Frequency range tuning and frequency interlock. Tests shall be performed to show that each filter section correctly tunes throughout the frequency range. At least 20 frequencies shall be selected throughout the frequency range for testing. Four frequencies (two pairs) shall be chosen to be within ±3 MHz of each other to verify the interlock functions. A radio set as specified in 3.6.11 shall be utilized to show remote control capability.

4.5.4 VSWR test. VSWR tests shall be conducted on all filter sections within the equipment for three frequencies (225.000 MHz, 300.000 MHz, and 399.975 MHz), as a minimum.

4.5.5 Insertion loss test. Tests to measure insertion loss shall be made at 225.000 MHz, 300.000 MHz, and 399.975 MHz.

4.5.6 Selectivity test. Selectivity tests shall be performed at 225.000 MHz, 300.000 MHz, and 399.975 MHz on each filter (section of the equipment).

4.5.7 EMI tests. EMI emission, susceptibility, and self-compatibility tests shall be performed in accordance with MIL-STD-462 as specified in 4.5.7.1 through 4.5.7.3.

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4.5.7.1 First article inspection. For first article inspection, all the specified requirements shall apply.

4.5.7.2 Quality conformance inspection. For quality conformance inspection, the requirements specified in a through c shall apply:

a. CE03
b. CE06
c. RE02

4.5.7.3 EMP tests. Conformance to the requirements of 3.6.12.2 shall be verified as specified in a through f:

a. Applicable test frequencies. Tests shall be performed for the injection frequencies of 0.01 MHz, 0.1 MHz, 0.5 MHz, 1 MHz, 2 MHz, 5 MHz, 10 MHz, 20 MHz, 50 MHz, and 100 MHz. In addition, the tests shall be performed at the test sample's critical frequencies, such as local oscillator, power switching frequencies and harmonics, RF frequencies, clock frequencies, bandpass frequencies, and so forth, as specified in the EMI test plan.

b. Apparatus. Test apparatus shall consist of the items specified in 1 through 4:

1. Source injection pulse generator with the capabilities specified in A through D:

A. Damped sinusoidal transient as shown in the applicable limit and as described by the equations specified in (1) and (2):

(1) $I_{pin}(t) = 1.05\ I(f)\ e^{-0.2ft}\ SIN\ (2\pi ft)$

(2) $V_{pin}(t) = 1.05\ V(f)\ e^{-0.2ft}\ SIN\ (2\pi ft)$

where:

$I_{pin}(t)$ = Pin to case current, in A at time, t
$V_{pin}(t)$ = Pin to case voltage, in V at time, t
f = Specific test frequency, in MHz
t = Time, in seconds
I(f) = Current determined from applicable limit curve of FIGURE 1
V(f) = Voltage determined from applicable limit curve of FIGURE 1

B. Variable amplitude so the injected transient can be increased from a minimum to a maximum specified level (that is, at least 10 percent, 50 percent, and 100 percent of the amplitude must be tested at each specified test frequency)

C. Frequency range capable of generating the required test frequencies

D. Current output level of at least 10 A into a 100-ohm load

2. Current probe
3. Oscilloscope with a minimum bandwidth capability of 100 MHz
4. Interconnecting test cable

c. Test setup and procedure. The test setup shall be as specified in d and e. A functional analysis shall be performed before the certification testing to identify the items specified in 1 through 3:

1. The interface pin connectors
2. The test sample functions which must be monitored to determine system failure or transient upset
3. The additional test frequencies (that is, interface cable resonant frequency, clock frequencies, data rates, and so forth)

d. Test setup. The test setup shall be as shown in FIGURE 2.

e. Test procedure. The test procedure shall be as specified in 1 through 5:

1. Apply the test signal to each interface pin of the test sample at each test frequency.
2. Monitor the test sample functions in all modes of operation to determine whether there is a failure or transient upset.
3. Inject one transient pulse on each interface pin at each test frequency.
4. The test sample shall be tested with its power off and on.
5. Record test frequency and level at which a failure or upset occurs.

f. Notes:

1. The test sample shall be tested on a ground plane. There shall be no insulated materials between the test setup and the ground plane.
2. The test cable running between the test sample and damped sinusoidal generator shall be kept 4 cm to 6 cm above the ground plane.

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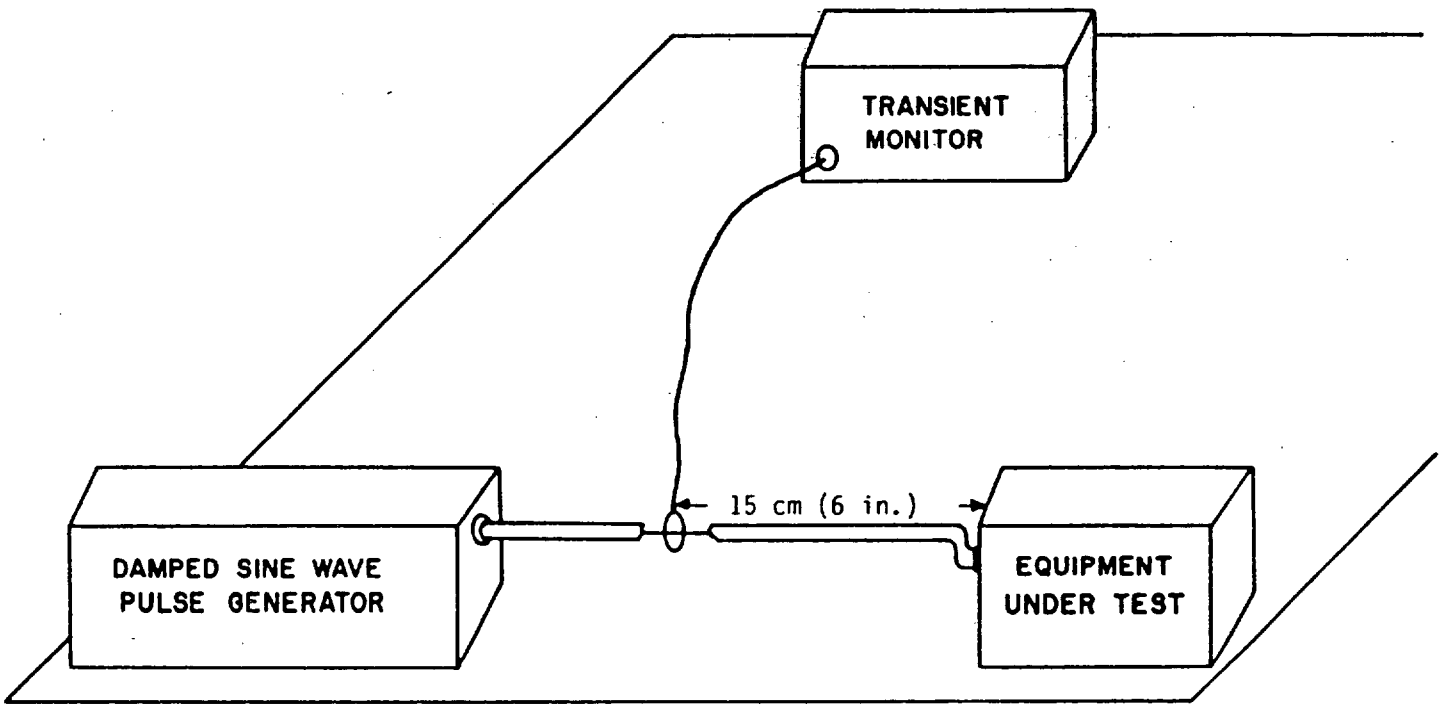


FIGURE 2. EMP test setup.

4.5.8 Leakage current test. The leakage current test shall be as specified in 4.5.8.1 and 4.5.8.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE MULTICOUPLER DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.8.1 Multicoupler connections. Each multicoupler directly connected to an external power source and units deriving power from the multicoupler shall be placed on an insulated surface. All safety ground conductors between the multicoupler and units deriving power from the multicoupler shall be intact. The safety ground conductor between the multicoupler under test and the source power shall be opened during the test. The multicoupler shall be connected as shown in FIGURE 3 and FIGURE 4 if it is connected to single-phase or DC source power. When DC measurements are performed do not reverse polarity unless the multicoupler has reverse polarity protection.

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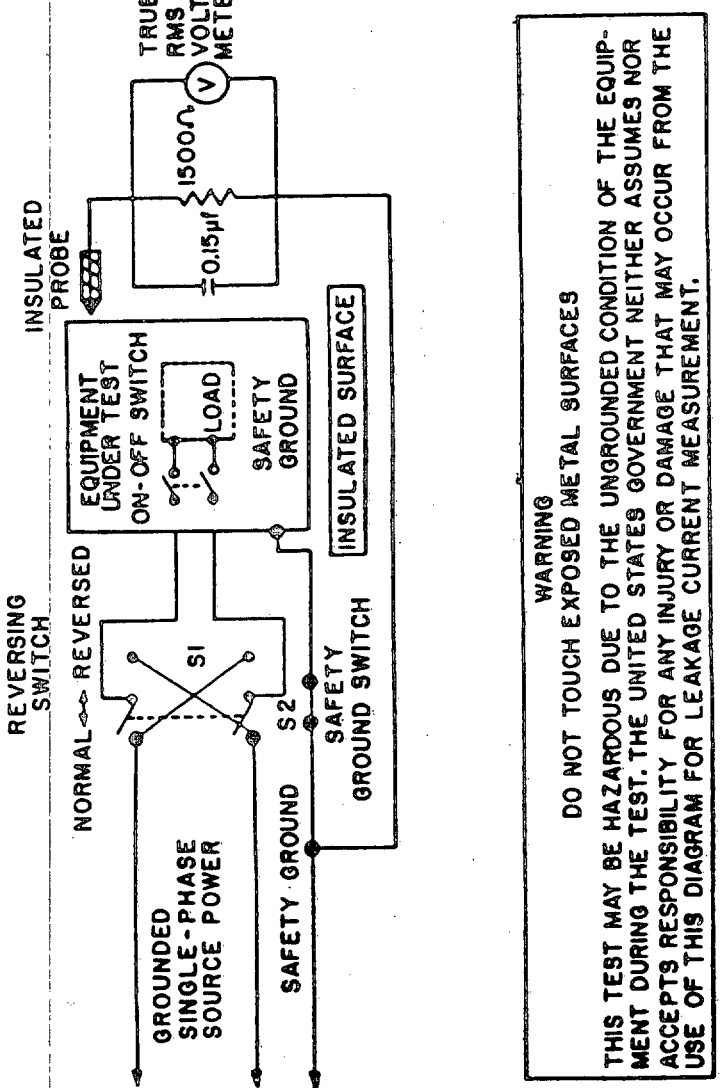


FIGURE 3. Single-phase test diagram for leakage current measurement.

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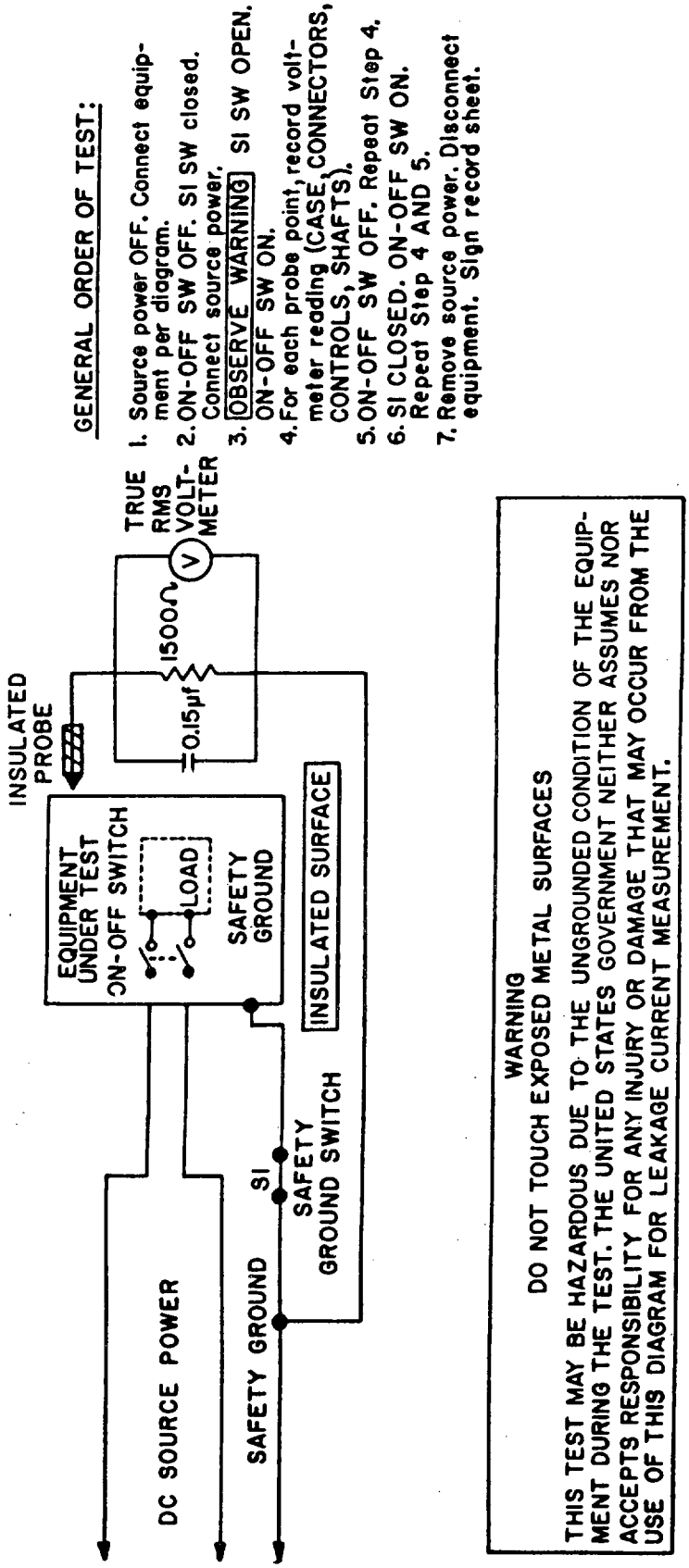


FIGURE 4. DC test diagram for leakage current measurement.

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4.5.8.2 Measurement. Leakage current shall be measured on the multicoupler in its normal operating configuration. Controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the multicoupler is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURE 3 and FIGURE 4. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.9 Input power interface. Input power shall be measured in accordance with the Power and power factor paragraph of MIL-E-16400. The multicoupler shall be subjected to the electrical tests of MIL-E-16400 to demonstrate compliance with the requirements of DoD-STD-1399, Section 300.

4.5.10 Operating test. The multicoupler, assembled to include all parts required for delivery of end items, shall be energized and subjected to an operating test to ensure qualitatively the correct functioning of the multicoupler including all external operating controls and exercise of all relays.

4.6 Reliability qualification test. A sample of at least three multicouplers shall be subjected to reliability qualification testing in accordance with Test Plan IXVII of MIL-STD-781. Environmental and operational duty cycles shall be as specified herein.

4.6.1 Reliability test temperature. The temperature cycle shall be between 0°C (+3, -0°C) and 50°C (+0, -3°C) so that 3 hours are spent at 50°C and 3 hours are spent at 0°C during each cycle. The time required to change from 0°C to 50°C shall be the minimum time for which the chamber is capable.

4.6.2 Reliability test vibration. Sine sweep shall be 4 Hz-33 Hz-4 Hz at 0.02 in. double amplitude. The time to traverse the spectrum up and down shall be 10 minutes.

4.6.3 Input voltage. Input voltage to all test samples shall be 115 VAC, 60 Hz.

4.6.4 Operation duty cycle. The multicouplers shall be operated continuously during the reliability test.

4.6.4.1 Tuning cycle. The tuning cycle of each multicoupler channel shall be in accordance with TABLE III. The multicoupler shall operate continuously in the transmit mode during the reliability qualification test.

TABLE III. Tuning cycle.

Hours	Channel 1	Channel 2	Channel 3	Channel 4
0 to 4	F1 1/	F2 2/	F3 3/	F4 4/
4 to 8	F3	F2	F1	F4
8 to 12	F3	F4	F1	F2
12 to 16	F4	F1	F2	F3
16 to 20	F2	F1	F4	F3
20 to 24	F2	F3	F4	F1

1/ F1 = 225.000 MHz
2/ F2 = 283.325 MHz
3/ F3 = 341.650 MHz
4/ F4 = 399.975 MHz

4.6.5 Performance measurement. At least daily, performance measurements shall be made to verify the performance characteristics specified in a through d:

a. Remote tuning
b. Tuning time
c. Insertion loss
d. Selectivity

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4.6.6 Failure. The procuring activity shall be notified of any failure during reliability testing. Classification of failures shall be made by the procuring activity in accordance with the Failure actions paragraph of MIL-STD-781. When two or more replaceable subassemblies are failed simultaneously, each shall be counted as a separate failure, even though one or more of the failures may be dependent.

4.6.7 Rejected equipment. If multicouplers are rejected as a result of reliability testing, the contractor shall immediately notify the procuring activity of the cause(s) for such rejection and propose a plan for correction of the deficiencies. When such corrections have been approved by the procuring activity and incorporated in the test units, the reliability qualification test shall be reperformed as specified herein.

4.7 Maintainability demonstration. The maintainability demonstration specified in 4.7.1 through 4.7.8 shall be performed on one multicoupler prior to first delivery under the contract.

4.7.1 Maintainability equipment demonstration. A maintainability equipment and BITE diagnostics demonstration shall be performed to verify conformance with the equipment M_{ct} BITE diagnostics requirements specified in 3.13.1 and 3.13.2.

4.7.1.1 Maintainability demonstration conditions. The multicoupler maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PCBs) and chassis-mounted electronic, electrical, electromechanical, and mechanical components of parts at the organizational level. One hundred candidate corrective maintenance tasks shall be determined in accordance with APPENDIX A of MIL-STD-471. The information specified in a through c shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

Faults of BIT functions shall be included as corrective maintenance tasks for the maintainability demonstration. The procuring activity or its authorized representative shall use the candidate tasks as a guide to determine a sample of 50 tasks for the demonstration.

4.7.1.2 Subassembly (module) maintainability. The module maintainability requirements shall be demonstrated by replacement of piece parts at the organizational, intermediate, or depot-level, as appropriate. A minimum of five faults shall be selected for each repairable module. Fault selection and simulation for each repairable module shall be in accordance with the procedure for the equipment maintainability demonstration. A total of 50 faults shall be determined by the procuring activity for the module maintainability demonstration.

4.7.2 Accept or reject criteria. Accept or reject criteria shall be as specified in 4.7.2.1 and 4.7.2.2.

4.7.2.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as specified in TABLE IV.

TABLE IV. Acceptance criteria.

	M_{ct}	M_{maxct}
Acceptance level	20	0
Sample size	50	50

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index (M_{ct}, M_{maxct}) is less than or equal to that shown in TABLE IV corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and determined as greater than or lesser than each index. The actual value shall also be determined. An accept decision can only be made when an

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accept decision is made for both M_{ct} and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability of being able to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision.

4.7.2.2 BITE diagnostics. At the insertion of each simulated fault or malfunction, the BITE diagnostics shall be exercised and compliance or noncompliance with the requirements for fault detection and isolation shall be determined. The accept criteria specified in a and b for fault detection isolation shall be not less than:

a. 48 out of 50 of the failures are detected
b. 42 out of 50 of the detected failures are isolated to the sole subassembly

More than one erroneous failure indication shall constitute failure of the demonstration.

4.7.3 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the multicoupler shall not interfere with on-line opration.

4.7.4 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the multicoupler will be installed.

4.7.5 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration.

4.7.6 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the multicoupler.

4.7.7 Rejection. Failure to conform to any of the maintainability requirements shall be cause for rejection of the maintainability demonstration.

4.7.8 Acceptance. The maintainability demonstration shall exhibit that the multicoupler conforms to the accept criteria.

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 through 4.8.2.2.

4.8.1 General workmanship. The multicouplers, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements specified in 3.14.1. Examination shall be made for finish in accordance with 3.5.4.1.

4.8.2 Workmanship screen. Vibration and temperature cycling shall be performed on each multicoupler. Vibration shall be performed prior to temperature cycling. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.8.2.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 5. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analyses of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the rms acceleration level. The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

4.8.2.2 Temperature cycling. Each multicoupler shall be subjected to 10 cycles of the temperature curve shown in FIGURE 6. The temperature rate of change shall be not less than 5°C per minute. Multicoupler power shall be turned on and off at the indicated times. The multicoupler shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the multicoupler comes within 10°C of the chamber temperature. When failures occur, the multicoupler shall be reworked and the cycling continued for a cumulative total of 10 cycles.

4.9 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

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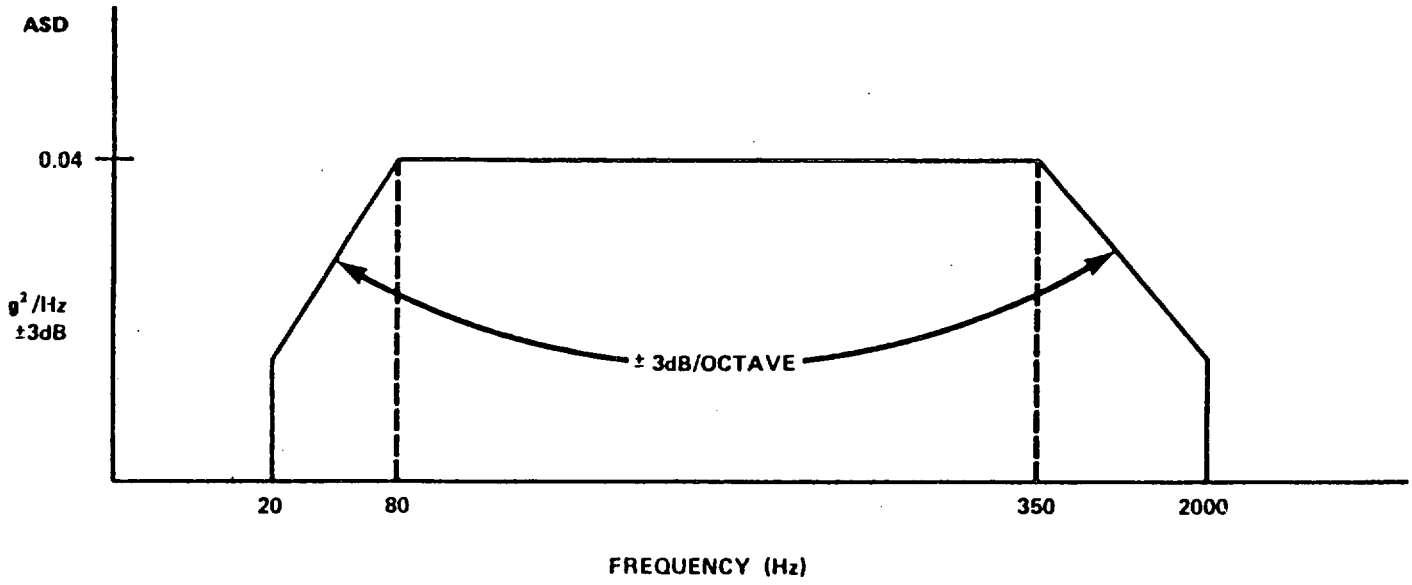


FIGURE 5. Random vibration curve.

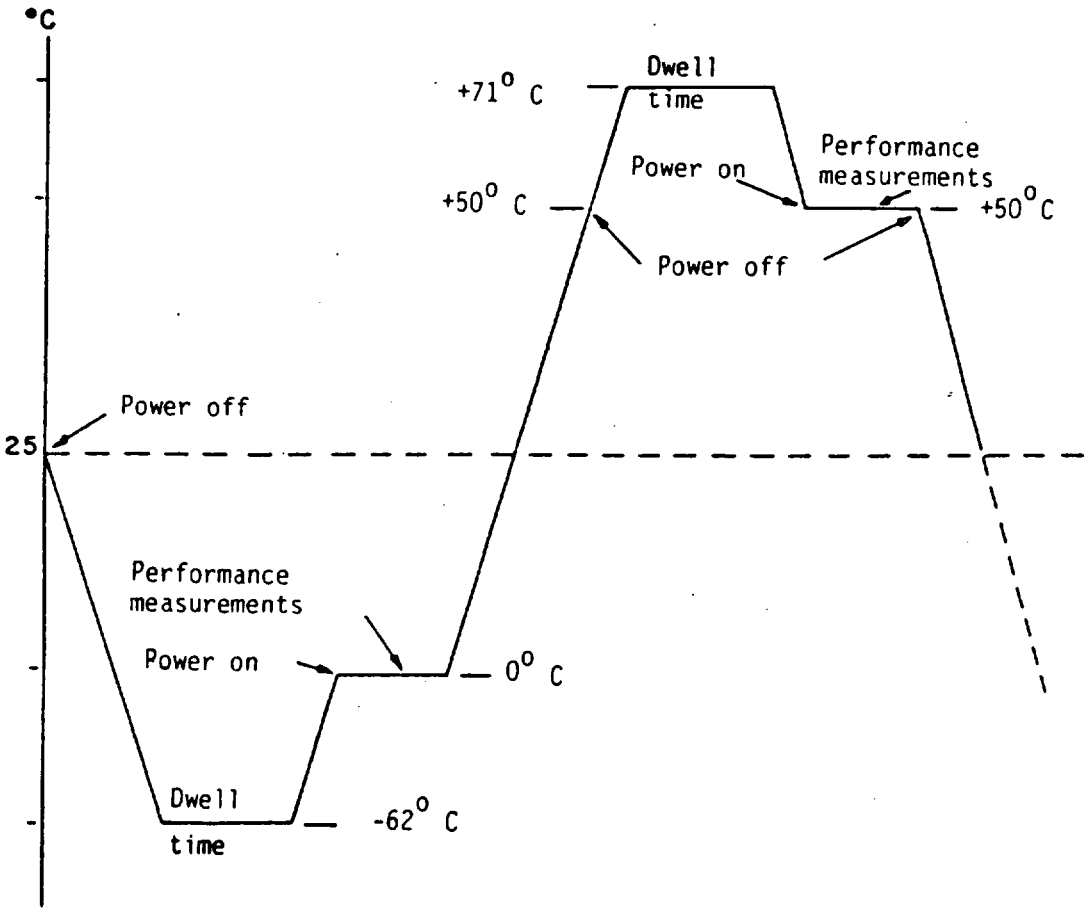


FIGURE 6. One cycle of temperature curve (not to scale).

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The multicoupler covered by this specification is intended for shipboard use.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition requirements should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than as specified in 4.3
c. Adjustment of the production control inspection sampling plan for compatibility with end item delivery requirements (see 4.4.2)
d. When reinspected environmental sample units may not be accepted (see 4.4.4)
e. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List, incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specifiction is cited in the following paragraph:

Paragraph	Data requirement	Applicable DID
4.4	Procedures, test	UDI-T-22710B

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. Definitions of terms used in this specifiction are given in 6.4.1 through 6.4.9.

6.4.1 Off-line test. Off-line test is a test performed while the tested equipment is not performing its normal operational function.

6.4.2 On-line test. On-line test is a test which is performed while the equipment is in normal operational use of the supported equipment.

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6.4.3 BITE. BITE is any device which is functionally separate from, but permanently connected to, the prime equipment and used for the express purpose of testing the prime equipment.

6.4.4 BIT. BIT test devices are devices which are an integral part of the equipment being tested. BIT may be automatic, manual, on-line, off-line, or a combination thereof.

6.4.5 Failure. The term failure refers to malfunction or parameter deviation that prevents the equipment from performing full capability in all modes, in the specified operational environment, to operating parameter levels set forth herein.

6.4.6 Subassembly. A subassembly is two or more parts which form a portion of an assembly, but have parts which are individually replaceable (such as, modules and PCBs with components).

6.4.7 Assembly. An assembly is a number of parts or subassemblies or any combination thereof, joined together to perform a specific function and capable of disassembly (such as, audio oscillator, amplifier, synthesizer, crystal oscillator).

6.4.8 Unit. An assembly or any combination of parts, subassemblies, and assemblies mounted together, normally capable of independent operation in a variety of situations is a unit.

6.4.9 System. All possible combinations of units required to perform a function is a system.

6.5 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.4.3).

6.6 Release for shipment. Release for shipment shall be as specified in 6.6.1 and 6.6.2.

6.6.1 Initial shipments. Initial shipment of end items shall be contingent upon successful completion of the environmental inspection (see 4.4.3).

6.6.2 Subsequent shipments. End item equipments, exceeding successive quantities required for an environmental sample, shall not be released for shipment prior to successful completion of the environmental inspection (see 4.4.3).

6.7 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY - EC

(Project No. 5985-N527)

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APPENDIX A

COUPLER GROUP, ANTENNA, OA-9123/SRC( )

10. SCOPE

10.1 Scope. This APPENDIX covers the functional description and the method of approach in the design of the OA-9123/SRC( ) antenna coupler group.

20. APPLICABLE DOCUMENTS

This section is not applicable to the APPENDIX.

30. REQUIREMENTS

30.1 Description. The OA-9123/SRC( ) is a shipboard tuned antenna coupler group used to couple a group of transmitters, receivers, transceivers, and combination thereof, to four separate antennas for operation in the frequency range of 225.000 MHz to 399.975 MHz, and used primarily by air controlling ships.

30.2 Functional description. The function of the OA-9123/SRC( ) is different from the basic OA-9123/SRC only, in that, the OA-9123/SRC( ) allows any of the four radios connected to it to tune to any frequency in the ultra-high frequency (UHF) band (225.000 MHz to 399.975 MHz), and receive or transmit without regard for what any of the other three radios are doing. The radios may be ON or OFF, transmitting or receiving, and they may be tuned to any frequency in the UHF band. (The basic OA-9123/SRC allows any of the four radios to transmit no closer than 3 MHz from the tuned frequency of any of the other three radios.)

30.3 Method of approach. The 3 MHz frequency limitation in the basic OA-9123/SRC is overcome through a software modification in the control head power supply section. A hardware modification removes the combiner, and, through appropriate impedance matching (50 ohms), outputs the four separate filters to four separate RF connectors for four separate antennas. The control of the entire multicoupler is still exercised by the control head power supply section.

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APPENDIX B

DERATING CRITERIA FOR ELECTRICAL AND ELECTRONIC PARTS

10. SCOPE

10.1 Scope. This APPENDIX covers the derating criteria for electrical and electronic parts.

20. APPLICABLE DOCUMENTS

This section is not applicable to the APPENDIX.

30. REQUIREMENTS

30.1 Derating. Derating is increasing the ratio (margin of safety) between part design limits and the allowable applied stresses (electrical, thermal, mechanical, and other environmental stresses). Derating can be accomplished by either selecting a part with broader limits or by limiting the applied stresses. Derating provides added protection for part variances and decreases the part degradation rate, thus increasing the expected life of a part. The following criteria provide basic information for the derating of electrical and electronic parts and indicate the maximum allowable application stress values to be considered during design. Since the safety margins of a part to both the threshold and time dependent stresses are generally unknown, parts should be derated to the maximum extent possible consistent with good design practices. When derating, the designer shall consider the part manufacturer's specified environmental and operating condition rating versus the actual environmental and operating conditions of the application and then apply derating factors so as not to exceed the maximum allowable application stresses herein.

Example: A wirewound resistor, RW67U--, which is rated 6.5 W at 25°C, linearly derating to zero W at 275°C, being operated in an equipment ambient of 70°C.

a. Compute rated power at 70°C;

$$P_{70\ \text{rated}} = P_{25\ \text{rated}} \times \frac{T_{\text{maximum}} - T_{\text{operating}}}{T_{\text{maximum}} - T_{\text{rating}}}$$

where $T_{\text{operating}} \times T_{\text{rating}}$

$$= 6.5 \times \frac{275 - 70}{275 - 25} = 6.5 \times \frac{205}{250}$$

$$= 5.33\ \text{W}$$

b. Apply derating criteria: from tables - limiting power to 0.50 maximum
P = 5.33 x 50 percent = 2.6 W

Parts which exceed a threshold of 80 percent of the maximum allowable application stress shall not be used prior to the approval of the procuring activity. Parts which equal or exceed the maximum allowable application stress at the highest electrical stress level, with the rationale for the specific application, shall be approved by the procuring activity.

Part type	Critical stress	Maximum allowable application stress	Remarks
MICROCIRCUITS			
Digital	Output current	0.80	
	Operating frequency	0.75	
Linear	Bias voltage	0.70	
	Input signal voltage	0.70	
	Output current	0.75	
	Operating frequency	0.75	
Voltage	Input voltage (rated maximum)	0.80	
	Output current (rated maximum)	0.75	
	Power dissipation (rated maximum)	0.60	
All	Junction temperature	110°C maximum	
All	Ambient temperature	85°C maximum	

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Part type	Critical stress	Maximum allowable application stress	Remarks
TRANSISTORS	See 1/		
General purpose	Power	0.50	
	Current	0.75	
	Voltage	0.75	
Power	Power	0.50	
	Current	0.75	
	Voltage	0.75	
Switching	Power	0.50	
	Current	0.75	
	Voltage	0.75	
All	Junction temperature	110°C maximum	
	Leakage current	See remarks	Consider leakage current to be at least twice rated value.
	Frequency	See 2/	
DIODES			
General purpose	Power	0.50	
	Peak inverse voltage (PIV)	0.50	
	Surge current	0.50	
	Forward (FWD) current	0.50	
Rectifier	Power	0.30	
	PIV	0.50	
	Surge current	0.50	
	FWD current	0.50	
Switching	Power	0.30	
	PIV	0.50	
	Surge current	0.50	
	FWD current	0.50	
Silicon controlled rectifier	Power	0.30	
	PIV	0.50	
	Surge current	0.50	
	FWD current	0.50	
Varactor	Power	0.50	
	PIV	0.75	
	FWD current	0.75	
Zener	Power	0.50	
	FWD current	0.50	
	Zener current	$0.5\ I_{Z_{maximum}} + I_{Z_{nominal}}$	
Reference	Power	0.30	
	FWD current	0.50	
	Zener current	$0.5\ I_{Z_{maximum}} + I_{Z_{nominal}}$	
All	Junction temperature	110°C maximum	

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Part type	Critical stress	Maximum allowable application stress 3/	Remarks
CAPACITORS FIXED	See 4/		
Ceramic	Voltage current AC/ripple Surge	0.50 0.70 0.70	
Mica, dipped	Voltage current AC/ripple Surge	0.60 0.70 0.70	
Glass	Voltage current AC/ripple Surge	0.50 0.70 0.70	
Porcelain	Voltage current AC/ripple Surge	0.50 0.70 0.70	
Paper	Voltage current AC/ripple Surge	0.50 0.70 0.70	Good low frequency characteristics
Plastic	Voltage current AC/ripple Surge	0.50 0.70 0.70	Mylar good to 1 MHz, polytetrafluoroethylene, polystyrene, and polycarbonate good to 1 GHz
Tantalum, foil or film	Voltage current AC/ripple Surge	0.50 0.70 0.70	Limit to 70°C ambient temperature
Tantalum, solid, dry	Voltage current AC/ripple Surge protection	0.50 0.70 3 ohms per V	Limit to 85°C ambient temperature
Tantalum, solid, wet	Voltage current AC/ripple	See remarks See remarks	Use of tantalum wet slug capacitors is prohibited except by prior approval of the procuring activity. 5/
ADJUSTABLE			
Ceramic	Voltage current AC/ripple Surge	0.50 0.70 0.70	
Glass	Voltage current AC/ripple Surge	0.50 0.70 0.70	
Air	Voltage	0.30	
All, fixed and adjustable	Leakage	See remarks	For all applications, consider leakage current as twice rated maximum value.

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Part type	Critical stress	Maximum allowable application stress 3/	Remarks
RESISTORS FIXED	See 6/		
Carbon composition	Power	0.50	
Insulated film	Power	0.50	
Wirewound precision			
1.0 percent total	Power	0.50	
0.1 percent total	Power	0.25	
Wirewound power	Power	0.50	
Thermistor	Power	0.50	Uses in other than zero power applications shall specify minimum wattage for the application.
ADJUSTABLE	See 6/		
Wirewound	Rated current	0.70	See 7/
Nonwirewound	Rated current	0.70	See 7/
All	Voltage	0.80	Maximum voltage shall not exceed 80 percent of the MIL-ratings.
RELAYS AND SWITCHES			
LOAD TYPE			
Capacitive	Contact current	0.75	Peak inrush current shall not exceed derated limit.
Resistive	Contact current	0.75	
Inductive	Contact current	0.40	
Motor	Contact current	0.20	
Filament	Contact current	0.10	
MISCELLANEOUS			
Transformer and chokes	Flux saturation level	0.75	Parts shall be used at winding temperature at least 30°C below rating.
	Insulation breakdown voltage	0.50	Maximum voltage between windings or between winding and shield shall not exceed 60 percent of rating for steady-state conditions or 90 percent of rating with transients.
	Current	0.70	

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Part type	Critical stress	Maximum allowable application stress	Remarks
Connectors	Current, pin	See remarks	Current derating for connector pins shall be in accordance with criteria for relays and switches, depending on type of loading. For paralleled pins, sum of current must not be more than 100 percent rated of lowest rated pin.
	Insulation breakdown voltage	0.50	

1/ Worst-case combination of DC, AC, and transient voltages shall be no greater than derated limit.
2/ Transistors rated at 500 milliwatts (mW) or less in TO-5 or TO-18 packages using one mil or smaller aluminum lead wires with thermocompression bonds shall not be used at repetition rates of less than 30 Hz. At 30 Hz and 25°C or less, derate the power dissipation to .20 and limit the input current to 50 mA. Above 25°C, derate an additional 1.33 mW per degree C. Gold lead devices should be used at repetition rates less than 30 Hz.
3/ Manufacturer's derating factors shall be applied before applying these factors.
4/ Voltage equals instantaneous total of DC, AC, surge, and transient voltages.
5/ When approved for use, tantalum wet slug capacitors shall be subjected to special screens such as temperature cycling and seal leakage.
6/ High density packaging may require further derating if ambient temperatures are increased.
7/ Rated current is defined as:

$$I_R = \sqrt{\frac{P_{maximum}}{R_{maximum}}}$$

And by limiting the current to 0.70 rated current, power is limited to 0.50 maximum power.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-28858A	Coupler Group, Antenna OA-9123/SRC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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